Supplementary Information for
An attractor account of numerosity perception

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Eudald Correig-Fraga[1]

This document accompanies the article *An attractor account of numerosity perception*. Supplementary Material 1 details the trial-level fitting; Supplementary Material 2 reports the full model comparison; Supplementary Material 3 establishes trial-level parameter recovery; and Supplementary Material 4 reports further model predictions and the individual-differences analysis.

Supplementary Material 1: Trial-level fitting

We reconstructed one row per trial from the openly available data of Cheyette and Piantadosi (2020). For each trial, we retained the presented numerosity, typed response, exposure duration, and response time. The sample size reported in the source article and the number of sessions in the open file refer to different samples. The article analysed four between-experiment arms with $N = 100$ in each arm, after excluding the ten highest-error participants per arm. The `DURATION` file used here instead pools the three duration-control arms (`NONE`, `AREA`, and `DENSITY`) without that exclusion. It contains 379 unique session identifiers that contributed at least one duration trial; 360 of these sessions contain data at all five exposure durations. Across the five durations (40, 80, 160, 320, and 640 ms) and numerosities $n = 1, \ldots, 15$, the reconstructed dataset contains 52,995 trials, approximately 10,600 per duration. All likelihoods are evaluated on the individual integer reports; no trials are aggregated into summary curves.

The model starts with the completed sensory estimate $y_0 \sim \mathcal{N}\big(n,\, \sigma(n)^2\big)$, with $\sigma(n) = \sqrt{s_{00}^2 + s_0\, n^{\lambda_0}}$ and fixed baseline $s_{00} = 0.05$; only s_0 and λ_0 are fitted (per duration in the joint fit). Integer memory components sit at $m = 1, \ldots, 19$ (the true range 1–15 plus four units of padding), and the readout softmax and the energy V are defined over this same support. Typed responses run up to $R_{\max} = 99$; a few per cent fall outside the well support (3.9–6.1% across durations) and are handled by the lapse channel below.

From $x = y_0$ the state follows the discrete descent $x \leftarrow x - \mathrm{clip}(\eta\, \partial V/\partial x,\, \pm\frac{1}{2} \min_m \sigma_m)$ with fixed step size $\eta = 1$ and global clamp $\frac{1}{2} \min_m \sigma_m$ (equal to $\frac{1}{2}\sigma_c$ for the power-law wells). Descent stops at the first zero-based iteration t for which either $|\mathrm{step}| < \mathrm{tol}$ with $\mathrm{tol} = 10^{-4}$, or the step reverses sign relative to the previous update (oscillation criterion: $\mathrm{step}_t \cdot \mathrm{step}_{t-1} < 0$). In both cases the reversing or sub-threshold step is *not* taken, so the resting state x^* is the state at detection; the stopping iteration is nevertheless counted, giving the recorded count $T = t + 1$ under the code's zero-based index. On the flat large-numerosity landscape, some trajectories do not meet the numerical settling criterion within an iteration budget compatible with the observed response times. Since reaching the eventual settling point can imply implausibly long predicted response times, we stop clean-up after G steps. Across the tested budget grid, $G \in \{40, 60, 100, 120, 140, 160, 250\}$, the response likelihood was highest at $G = 120$ ($\Delta \log L = 0$), which we use in the clean-up model.

For each model and duration the response probability of Eq. (4) marginalizes the readout over the sensory channel. The integral is computed deterministically, so the objective is free of Monte-Carlo noise. Gauss–Hermite quadrature is inappropriate here: for the attractor models the clean-up map $y_0 \mapsto x^*$ is near-discontinuous (evidence snaps to basins), and Gauss–Hermite carries a 1.7–4% error even at 241 nodes. We therefore use a uniform standard-normal grid on $z \in [-7, 7]$ with Gaussian weights for every model, so the comparison is on identical footing. Against high-sample Monte-Carlo estimates, a grid of 801 nodes matches the integral to $\leq 2 \times 10^{-3}$ (mean 1.5×10^{-4}). The optimizer uses a cheaper grid (201 nodes); the reported log-likelihoods are recomputed at the fitted parameters on a fine grid (1601 nodes).

We never discard out-of-support responses: the final response probability is $p_{\mathrm{final}}(r \mid n) = (1 - \ell)\, p(r \mid n) + \ell \,/\, R_{\max}$ with $R_{\max} = 99$, a uniform contamination over $r = 1, \ldots, 99$ that places positive mass everywhere. The lapse is counted in every AIC and BIC. It is fitted separately at each duration in the attractor, static-readout, Weber/ANS, and C&P fits. The published-form joint C&P fit instead shares one lapse across durations; we therefore also fit a matched joint C&P variant with duration-specific lapses.

Response time is modelled as log-normal, $\log \mathrm{rt} \sim \mathcal{N}(\log(t_0 + m(n)), \sigma_{\mathrm{rt}})$, so main-text Eq. (3) is its location on the response-time scale—the predicted median and geometric mean, not the arithmetic mean. The metric $m(n)$

[1] Innovamat Education, Avinguda de la Generalitat, 217, Sant Cugat del Vallès, Catalonia (Spain)

Corresponding author:
Eudald Correig-Fraga, Innovamat Education, Avinguda de la Generalitat, 217, Sant Cugat del Vallès, Catalonia (Spain).
Email: eudald.correig@innovamat.com

Table S1. Fitted parameters of the joint winning model. Shared: $\sigma_c = 0.0185$, $p = 1.86$, $G = 120$, $s_{00} = 0.05$; selected RT link $\tau = 0.01144$, $\kappa = 0$, $t_0 = 1.30$ s (fixed). Sensory noise (s_0, λ_0), lapse ℓ, and RT residual σ_{rt} vary by duration (landscape held fixed for the RT fit). Well support $m = 1$–19; clean-up uses $\eta = 1$, $\mathrm{tol} = 10^{-4}$, and the oscillation stop above. We summarize the scale and exponent jointly through $n_{\mathrm{width}} = \sigma_c^{-1/p} = 8.5$.

Duration	s_0	λ_0	ℓ	σ_{rt}
40 ms	0.070	2.25	0.115	0.472
80 ms	0.0033	3.30	0.067	0.471
160 ms	0.0017	3.37	0.060	0.479
320 ms	0.00060	3.60	0.076	0.497
640 ms	0.00021	3.91	0.066	0.526

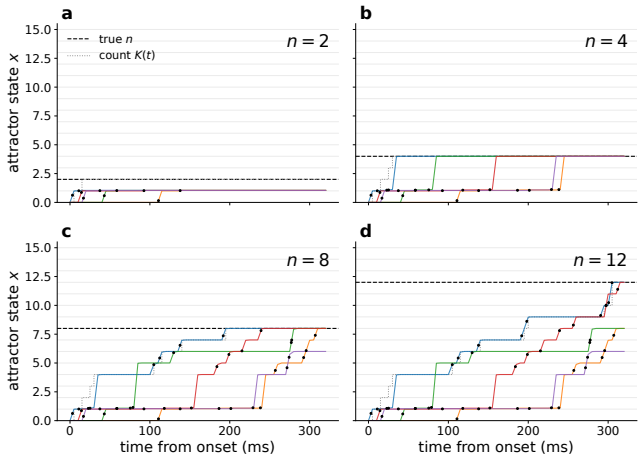


Figure S1. The purely driven model captures the wrong integer. State $x(t)$ of the purely driven model on the winning (sharp) landscape at 320 ms, for five example trials each at **a**, $n = 2$; **b**, $n = 4$; **c**, $n = 8$; **d**, $n = 12$. Black dots mark item registrations; the dotted line is one trial's running count $K(t)$; the dashed line is the true n. Driven by the accumulating count, the state is captured by whichever integer the partial count currently resembles and is dragged up through the wrong wells; for small n it is trapped near 1, and for large n it has not reached n by offset, with high across-trial variance. A partial count is not diagnostic of the total, so it is a poor retrieval cue.

is the expected posterior entropy ($\kappa\,\mathbb{E}[H]$), the expected convergence time ($\tau\,\mathbb{E}[T]$), or their sum (the hybrid). We fix the basal latency at $t_0 = 1.30$ s: because RT is timed from stimulus offset, this intercept absorbs mask-viewing, typing, and motor time, and at the longest exposures the geometric-mean RT for $n = 1$ approaches $\sim$1.30 s, placing t_0 at the floor of the observed chronometry. It is held fixed for all models and durations and is not estimated from the likelihood. Link coefficients are shared across durations and residual scatter σ_{rt} is profiled per duration. The steps-only link is selected by AIC, narrowly ahead of the hybrid ($\Delta\mathrm{AIC} = 0.24$) and far ahead of entropy alone ($\Delta\mathrm{AIC} = 1579.5$). All three are fitted with the landscape held fixed at its response-fit values, providing a constrained cross-channel test of the fitted landscape.

The deterministic objective is minimized by differential evolution (Storn and Price 1997) followed by a Nelder–Mead polish, with fixed seeds.

Supplementary Material 2: Model comparison and constrained fits

Table S2 shows the joint response-model fits and states the sharing configuration used for every row. In the power-law attractor and static fits, only the landscape parameters (σ_c, p) are shared; in the constant-width ablations, $p = 0$ is fixed and only σ_c is shared. The sensory parameters (s_0, λ_0) and lapse are duration-specific. The Weber/ANS model has no separable landscape, so its four parameters remain duration-specific.

The normative benchmark is the bounded-optimal model of Cheyette and Piantadosi (2020), reimplemented from their published code: a Zipf $(1/n^2)$ prior read out through an information-bottleneck posterior whose information budget is $\min(B, Rt)$ bits, with t the exposure duration. Its published joint form shares B, R, and one lapse across durations ($k = 3$). Because the other joint models use duration-specific lapses, the matched C&P comparison shares only B and R and fits five lapses ($k = 7$); this is the C&P row used in main-text Fig. 2a. Our implementation reproduces the published parameters on simulated data and matches the reference likelihood to machine precision.

Turning the clean-up dynamics on adds a single parameter, the shared give-up budget G (Supplementary Material 1); the dynamics are otherwise deterministic given the landscape. Even net of this penalty, the attractor improves AIC by 721.6 over its matched static readout in the joint fit. Fixing $p = 0$, and hence $\sigma_n = \sigma_c$, worsens AIC by 3488.2 (both models share the budget, so this comparison is unaffected). Clean-up therefore remains beneficial under the ablation, but it cannot compensate for removing magnitude-dependent broadening.

We also explored a further alternative that tries to model both the accumulation of numerical sensory input and inference as the same mechanism. This means that we dispense with the completed estimate altogether, so that the items are registered one at a time over the exposure and the running count itself drives the attractor, while the landscape pulls the state towards the stored integers (main-text Methods), with the response read out at offset. This version fits notably worse than the completed-estimate attractor (Table S2), providing evidence for a two-step process of numerical perception: accumulation and then readout. The reason is visible in the trajectories (Fig. S1): on the sharp small-number wells the state is captured by whatever integer the partial count currently resembles and is dragged up through the wrong wells as the count grows, because a partial count is not diagnostic of the total. A partial count is thus a poor retrieval cue, unlike a degraded but complete estimate.

The width exponent p is a free parameter of the model, shared across durations and fitted at $p = 1.86$ (Table S1). It grows the wells from narrow and well-separated at small n to wide and overlapping at large n, but p and the scale σ_c are not interpreted singly. We therefore summarize their joint effect with the continuous width-based index n_{width}, defined by $\sigma_n = 1$. What separates our account from the Weber/ANS magnitude code is the discrete pattern-completion architecture: at the identical trial-level convention the attractor beats the magnitude code

Table S2. Trial-level response-model comparison (joint shared-landscape fit). k is the total number of free parameters over all five durations. Information criteria use all 52,995 responses and are reported relative to the power-law attractor. The joint power-law models share σ_c and p; the constant-width ablations fix $p = 0$ and share only σ_c. In all cases, s_0, λ_0, and lapse are fitted per duration. The clean-up (attractor) models additionally count the shared give-up budget G (Supplementary Material 1) in k, so turning the clean-up on adds one parameter. Weber/ANS remains fully duration-specific. The C&P model shares B and R but uses a duration-specific lapse ($k = 7$).

Model	k	ΔAIC / ΔBIC
Power-law attractor	**18**	**0 / 0**
Driven (accumulated count)	11	7954 / 7891
Power-law readout (static)	17	722 / 713
Constant-width attractor ($p = 0$)	17	3488 / 3479
Constant-width readout (static; $p = 0$)	16	3718 / 3700
Weber/ANS	20	838 / 855
Normative (C&P 2020)	7	11453 / 11355

Table S3. Parameter recovery of the shared-landscape model. Identity R^2 and Spearman ρ between generating and recovered values. Shared quantities are evaluated across the 20 complete synthetic datasets. For duration-specific quantities, R^2 removes duration means and ρ ranks values within duration before pooling the 100 duration blocks. Scale parameters are compared on the log scale.

Quantity	N	R^2	Spearman ρ
σ_c (well scale)	20	0.980	0.988
p (width exponent)	20	0.997	0.998
s_0 (sensory scale)	100	0.538	0.788
λ_0 (sensory exponent)	100	0.926	0.929
ℓ (lapse)	100	0.964	0.976
n_{width} (derived)	20	0.998	0.995
σ_3 (well width at $n = 3$)	20	0.994	0.997
$\sigma(n = 5)$ (sensory sd)	100	0.991	0.982

by $\Delta\text{AIC} = 838$ (Table S2). Fitted sensory, lapse, and RT-link values for the joint winning model are reported in Table S1 (Supplementary Material 1).

The comparison shares one landscape across durations. The attractor beats the Weber/ANS fit by $\Delta\text{AIC} = 838$ ($\Delta\text{BIC} = 855$) and the matched C&P fit by $\Delta\text{AIC} = 11{,}453$ ($\Delta\text{BIC} = 11{,}355$).

Supplementary Material 3: Parameter recovery

To test identifiability under the shared-landscape model, we generated 20 complete synthetic datasets, each containing all five exposure durations. For each dataset, σ_c and p were jittered once and shared across durations, whereas s_0, λ_0, and lapse were jittered separately for each duration. Responses were sampled with the empirical trial count in every duration-by-numerosity cell, including the lapse channel, on a 1601-node quadrature grid. Each complete dataset was then refitted with the same categorical likelihood, 201-node fitting grid, and shared-landscape parameterization as the headline analysis. Thus, the landscape statistics contain 20 independent fits and the duration-specific statistics contain 100 fitted blocks. Table S3 reports fitted and derived quantities; Figure S2 shows their scatter and error correlations.

Sharing the landscape across all five durations makes its two fitted parameters identifiable at this sample size: identity R^2 is 0.980 for σ_c and 0.997 for p. Their errors nevertheless trade off ($\rho = -0.87$), so the two coefficients should not be assigned separate psychological meanings. The derived width profile is recovered more directly: $R^2 = 0.994$ for σ_3 and $R^2 = 0.998$ for n_{width}, whose median absolute error is 0.024 numerosities. The two sensory parameters show the analogous trade-off ($\rho = -0.94$): within duration, s_0 alone is only moderately recovered ($R^2 = 0.538$), whereas the combined sensory-noise profile $\sigma(n) = \sqrt{s_{00}^2 + s_0 n^{\lambda_0}}$ is recovered accurately ($R^2 = 0.991$ at $n = 5$). Lapse recovery is also strong ($R^2 = 0.964$). These results support interpreting the width and sensory profiles rather than assigning causal meaning to their component coefficients; n_{width} remains a width-based summary, not an estimate of where attractors disappear.

Supplementary Material 4: Model predictions and individual differences

Exact-accuracy and response time shoulders

The predicted proportion of exact responses shows a small shoulder near n_{width}, even though the fitted width law and sensory-noise profile vary smoothly with n (main-text Fig. 3b). The shoulder vanishes when clean-up is switched off at identical parameters (Fig. S3), so it reflects the dynamics on the integer lattice. The same occurs in response time around $n = 4$–8 (main-text Fig. 3c), although here we cannot compare with static models because they have no dynamics from which to derive response-time predictions. The discrepancy between these predicted shoulders and the behavioural data remains unexplained, but it may reflect heterogeneity in landscape geometry or stopping policies, as discussed in the main text.

Individual differences

We ask whether a sharper landscape goes with lower estimation noise across participants using one hierarchical trial-level model with non-overlapping response channels (main-text Fig. 2c). The analysis includes all 52,933 responses from the $N = 360$ sessions that completed the five exposure durations. Small-numerosity trials ($n \leq 8$) enter the power-law attractor likelihood with participant effect u_i^n on $\log n_{\text{width}}$; the width exponent and duration-specific sensory channels are fixed at the all-duration group estimates. Larger-numerosity trials ($n \geq 9$) enter a discretized linear Weber likelihood, $r^\star \sim \mathcal{N}(\gamma_d n, (wn)^2)$

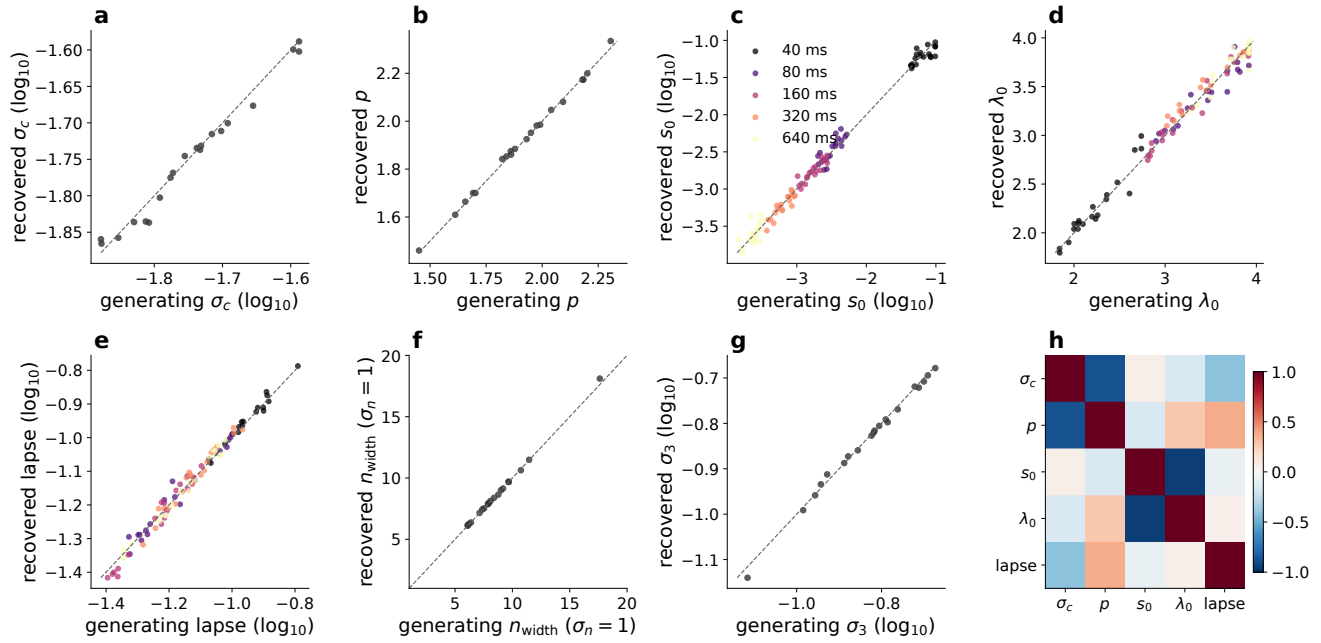


Figure S2. Parameter recovery of the shared-landscape model. Recovered versus generating values for the five fitted parameters and the derived n_{width} and σ_3; dashed lines are identity. Shared landscape quantities are shown once per complete dataset in black, whereas duration-specific quantities are coloured by exposure duration. The bottom-right panel is the Spearman correlation matrix of recovery errors, showing the two main trade-offs, σ_c–p and s_0–λ_0.

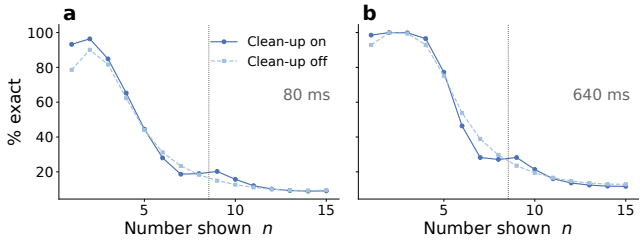


Figure S3. Clean-up generates an exact-accuracy shoulder. The shoulder is generated by the integer clean-up dynamics and disappears in the matched static control. It is more pronounced than in the behavioural data, making it both a distinctive prediction of the mechanism and a residual of the current homogeneous landscape. Predicted proportion of exact responses at 80 ms (**a**) and 640 ms (**b**) with clean-up on (solid) and off (dashed), at identical parameters; dotted verticals mark n_{width}. The shoulder occurs near this width-based proxy and disappears without clean-up.

on $r = 1, \ldots, 99$, with participant effect u_i^w on $\log w$ and pooled duration-specific gain γ_d.

Participant lapse is estimated in the same model, $\text{logit}(\ell_{i,d}) = \alpha_{\ell,d} + u_i^{\ell}$, with a duration-specific intercept and a participant effect shared across durations and channels. The joint Gaussian distribution of $(u_i^n, u_i^w, u_i^{\ell})$ is parameterized so that $\rho_{nw\cdot\ell}$ is the n_{width}–Weber correlation conditional on lapse. Tensor Gauss–Hermite marginalization uses nine nodes per dimension (729 joint nodes). Two dispersed starts converge to the same alternative optimum; the $\rho_{nw\cdot\ell} = 0$ likelihood-ratio null re-optimizes every nuisance parameter. Participant lapse covaries with both outcomes in the potentially confounding directions ($\hat{\rho}_{n\ell} = -.45$, $\hat{\rho}_{w\ell} = .59$), but the partial association remains $\hat{\rho}_{nw\cdot\ell} = -.627$ ($\chi^2(1) = 105.0$, $p = 1.2 \times 10^{-24}$). Main-text Fig. 2c plots duration-averaged empirical-Bayes estimates.

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